

# Dipanshu Barfa

Backend Engineer · Java · Spring Boot  
Spring AI · LLM Integrations

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## PROFESSIONAL SUMMARY

Backend Engineer with ~2 years of product-based startup experience building enterprise-grade systems with **Java & Spring Boot**. Shipped AI-powered email intelligence features used across web, mobile, and voice clients; designed an enterprise **Policy-as-Code platform** integrating Open Policy Agent for centralized RBAC/ABAC authorization at scale. Hands-on with **LLM pipelines (Groq)**, async Kafka architectures, multi-provider integrations (Microsoft Graph, Gmail, Exchange EWS), and DevSecOps tooling (Docker, Kubernetes, Trivy, Snyk). Proven ability to own features end-to-end in fast-moving teams while mentoring junior developers and contributing to architectural decisions.

## TECHNICAL SKILLS

|                                  |                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Backend</b>                   | Java, Spring Boot, Spring AI, REST APIs, AOP, Hibernate / JPA, Feign Client, Maven                             |
| <b>AI &amp; LLM</b>              | Spring AI, Groq, LLM APIs, RAG Architecture, LLM Prompt Engineering, Open Policy Agent (OPA / Rego)            |
| <b>Databases &amp; Messaging</b> | MariaDB, MySQL, PostgreSQL, Cassandra, Redis, Apache Kafka                                                     |
| <b>Integrations</b>              | Microsoft Graph API, Gmail API, Exchange EWS (multi-provider strategy pattern), Keycloak, API Gateway, Swagger |
| <b>DevOps &amp; Security</b>     | Docker, Kubernetes, Helm Charts, ArgoCD, SonarQube, Trivy, Snyk, Semgrep, FOSSA, SBOM, GitLab SAST             |
| <b>Observability &amp; Tools</b> | Apache SkyWalking, Melody APM, ConfigMap, Git / GitLab, IntelliJ, Cursor, VS Code                              |

## PROFESSIONAL EXPERIENCE

**VisionWaves Technologies Pvt Ltd** — Product-Based Startup, Indore  
**Backend Engineer**

Aug 2024 – Present

### Email Singularity — AI Email Intelligence Platform

Production · AI / LLM

**Stack:** Java, Spring Boot, Spring AI, Kafka, Redis, MariaDB, Microsoft Graph API, Gmail API, Exchange EWS, Groq, LLM APIs

- Built **15+ production REST APIs** powering mail summary, day-overview dashboards, inbox prioritization, and conversation threading — enabling users to triage inbox without reading every email.
- Developed **MailSummaryService** aggregating AI-enriched metadata (category, priority, action items, suggested replies) from MariaDB with live mailbox sync via Microsoft Graph, EWS, and Gmail APIs.
- Engineered multi-step **LLM pipeline** (Groq LLM API) for email categorization (FYI/Attention/Event), tone detection, priority scoring, and objective extraction across 25+ production prompts.
- Implemented **config-driven strategy pattern** for multi-provider backends (Graph, EWS, Gmail) enabling single-codebase deployment across Office 365, on-prem Exchange, and Google Workspace.
- Built **Kafka async pipeline** decoupling email ingestion from AI enrichment for horizontal scale; integrated S3 + RAG pipeline for document attachment Q&A on PDFs and Office files.
- Implemented paginated inbox APIs with folder filtering, date-range queries, thread-level aggregation; built calendar APIs for slot discovery and NLP-driven scheduling via LLM intent parsing.

### OPA Policy Manager & Governance SDK

Enterprise · Policy-as-Code

**Stack:** Java, Spring Boot, OPA / Rego, Kafka, MySQL, Hibernate Envers, Docker, Kubernetes, Feign, Caffeine Cache

- Developed **Governance SDK** — reusable Java library enabling Spring Boot microservices to enforce OPA-based authorization via @OpaAuthorize annotations with fail-closed 403 handling; adopted by 10+ service teams.
- Built **AOP-based request interceptor** forwarding HTTP context (token, headers, path, method, body) to OPA for real-time policy evaluation before business logic — zero boilerplate for consuming services.
- Implemented **Kafka-based Decision Records** capturing OPA input/output, before/after entity snapshots, and causation chains on every governed action for tamper-evident compliance audit trails.
- Contributed to **Policy Manager APIs**: lifecycle management (Draft → Approve → Deploy → Retire), OPA bundle generation, inline Rego testing, and multi-tenant dataset isolation via Hibernate filters & TSID IDs.

- DevOps & Observability:** Configured Docker, Kubernetes, Helm Charts, ConfigMaps, and ArgoCD pipelines; set up Apache SkyWalking and Melody APM for distributed tracing and performance monitoring.
- Security & Quality:** Resolved POM vulnerabilities via Trivy, Snyk, Semgrep, FOSSA, SBOM; maintained SonarQube quality gates and GitLab SAST security pipelines across all microservices.
- Collaboration & Leadership:** Mentored junior engineers, drove sprint discussions, and contributed ideas in architectural planning meetings.

## EDUCATION

### B.Tech — Computer Science & Engineering

2020 – 2024

Shri Vaishnav Vidyapeeth Vishwavidyalaya (SVVV), Indore, Madhya Pradesh · CGPA: 7.17 / 10